

8

CALENDARS

8.1. Calendars

Here you mainly deal in finding the day of the week on a particular given date.

The process of finding this depends on the number of odd days.

Odd days are quite different from the odd numbers.

- **Odd Days:** The days more than the complete number of weeks in a given period are called odd days.
- **Ordinary Year:** An year that has 365 days is called Ordinary Year.
- **Leap Year:** The year which is exactly divisible by 4 (except century) is called a leap year.

Example: 1968, 1972, 1984, 1988 and so on are the examples of Leap Years.

1986, 1990, 1994, 1998, and so on are the examples of non leap years.

Note: The Centuries divisible by 400 are leap years.

Important Points:

- An ordinary year has 365 days = 52 weeks and 1 odd day.
- A leap year has 366 days = 52 weeks and 2 odd days.
- Century = 76 Ordinary years + 24 Leap years.
- Century contain 5 odd days.
- 200 years contain 3 odd days.
- 300 years contain 1 odd day.
- 400 years contain 0 odd days.
- Last day of a century cannot be Tuesday, Thursday or Saturday.
- First day of a century must be Monday, Tuesday, Thursday or Saturday.

Explanation:

$$\begin{aligned} 100 \text{ years} &= 76 \text{ ordinary years} + 24 \text{ leap years} \\ &= 76 \text{ odd days} + 24 \times 2 \text{ odd days} \\ &= 124 \text{ odd days} = 17 \text{ weeks} + 5 \text{ days} \end{aligned}$$



- ∴ 100 years contain 5 odd days.
No. of odd days in first century = 5
- ∴ Last day of first century is Friday.
No. of odd days in two centuries = 3
- ∴ Wednesday is the last day.
No. of odd days in three centuries = 1
- ∴ Monday is the last day.
No. of odd days in four centuries = 0
- ∴ Sunday is the last day.

Since the order is continually kept in successive cycles, the last day of a century cannot be Tuesday, Thursday or Saturday.

So, the last day of a century should be Sunday, Monday, Wednesday or Friday.

Therefore, the first day of a century must be Monday, Tuesday, Thursday or Saturday.

8.2. Working Rules

Working rule to find the day of the week on a particular date when reference day is given:

Step 1: Find the net number of odd days for the period between the reference date and the given date (exclude the reference day but count the given date for counting the number of net odd days).

Step 2: The day of the week on the particular date is equal to the number of net odd days ahead of the reference day (if the reference day was before this date) but behind the reference day (if this date was behind the reference day).

Working rule to find the day of the week on a particular date when no reference day is given

Step 1: Count the net number of odd days on the given date

Step 2: Write:

For 0 odd days – Sunday

For 1 odd day – Monday

For 2 odd days – Tuesday

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For 6 odd days - Saturday

Examples: If 11th January 1997 was a Sunday then what day of the week was on 10th January 2000?

Solution: Total number of days between 11th January 1997 and 10th January 2000

$$= (365 - 11) \text{ in } 1997 + 365 \text{ in } 1998 + 365 \text{ in } 1999 + 10 \text{ days in } 2000$$



$= (50 \text{ weeks} + 4 \text{ odd days}) + (52 \text{ weeks} + 1 \text{ odd day}) + (52 \text{ weeks} + 1 \text{ odd day}) + (1 \text{ week} + 3 \text{ odd days})$

Total number of odd days $= 4 + 1 + 1 + 3 = 9 \text{ days} = 1 \text{ week} + 2 \text{ days}$

Hence, 10th January, 2000 would be 2 days ahead of Sunday i.e. it was on Tuesday.

Examples: What day of the week was on 10th June 2008?

Solution: 10th June 2008 = 2007 years + First 5 months up to May 2008 + 10 days of June

2000 years have 0 odd days.

Remaining 7 years has 1 leap year and 6 ordinary years $\Rightarrow 2 + 6 = 8 \text{ odd days}$

So, 2007 years have 8 odd days.

No. of odd days from 1st January 2008 to 31st May 2008 $= 3+1+3+2+3 = 12$

10 days of June has 3 odd days.

Total number of odd days $= 8+12+3 = 23$

23 odd days $= 3 \text{ weeks} + 2 \text{ odd days}$.

Hence, 10th June, 2008 was Tuesday.

